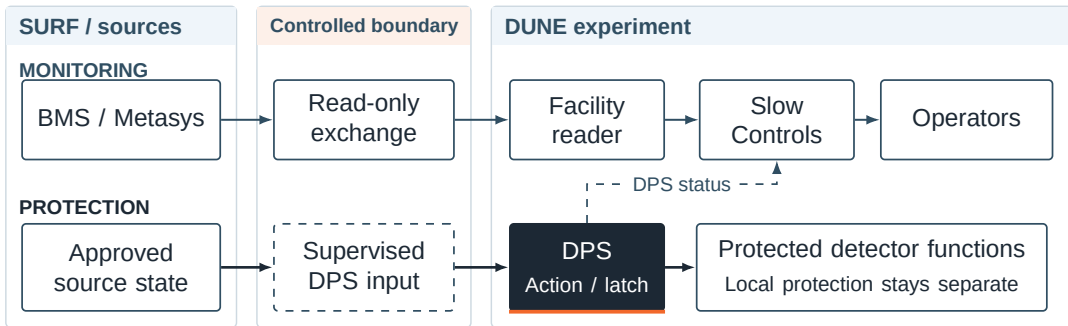


How facility signals reach DUNE

Proposed paths for monitoring and detector protection.



Functional view

Protection link to be defined.

Lost monitoring

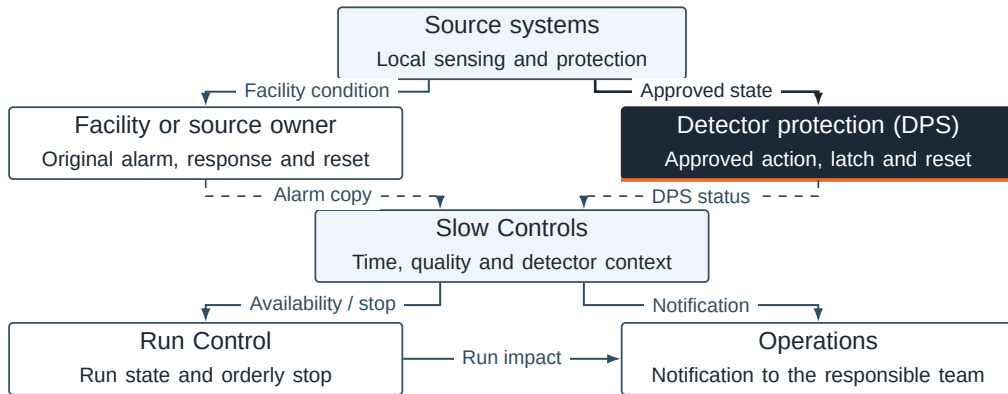
Missing or old data is visible.

Recovery

Restart stays with its owner.

Alarm flow and ownership

Information moves through the experiment; authority stays with the owner.



DUNE acknowledgment applies to the displayed copy, not the source alarm or DPS reset.

Facility conditions

The source definition and agreed cause and effect determine the response.

CONDITION	POSSIBLE DUNE RESPONSE
Power / reserve	An agreed request for an orderly DAQ stop.
Network / BMS loss	Stale or unknown data; the loss is visible.
Fire / low oxygen	Source response; detector action only where approved.
Cryogenic / LAr leak	Response follows an explicitly defined source condition.
Cooling / water	Local protection; agreed DPS action where applicable.
Weather / access	SURF operating restrictions guide the response.

Loss of monitoring is a design case for the independent protection path.

Proposed data exchange

Protocol support and source ownership remain to be confirmed.

INTERFACE	CANDIDATE	INFORMATION FLOW
BMS readout	BACnet/IP	Reader queries selected source points.
DUNE view	OPC UA	Reader publishes to Slow Controls.
Equipment health	SNMPv3	Supported power and network telemetry.

Read access exchanges requests and replies without granting control authority.

BACnet/SC is an option where supported. The DPS interface is defined separately.

From concept to agreed interface

Three areas for discussion with the source and experiment teams.

1 Point meaning and ownership

Each point has a defined source, state, quality and owner.

2 Data path and timing

The interface describes read access, update rate and data age.

3 Response and recovery

Checks cover alarm routing, loss, approved protection action and reset.
